THE UNIVERSITY OF AZAD JAMMU AND KASHMIR MUZAFFARABAD



Department of Software Engineering

Subject: Software Testing Lab

Lab Instructor: Engr. Fahad Nisar

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Lab Task:2

Name: Syed Jaffar Mukhtar
Roll No: 2024 – SET – 19

1. Introduction

This report presents application rules (acceptance criteria) and ten repeatable test cases for TodoMatic, a React-based task management application. The purpose of this lab task is to identify the expected behaviour of selected application features and design test cases using clear preconditions, test data, ordered steps, and observable expected results.

2. Objective

To define application rules and design ten clear, repeatable test cases covering valid and invalid inputs or actions.

3. Application Details

Field	Details
Application name	TodoMatic (MDN Todo React)
GitHub repository	https://github.com/mdn/todo-react
Package name	moz-todo-react
Package version shown in package.json	0.3.0
Technology	React 18.2.0 and Vite 5.4.21
Local development address	http://localhost:3000
Download method	GitHub ZIP archive
Commit information	Not available from the downloaded ZIP because the .git repository metadata was not included.

Version note: The value 0.3.0 is the package version recorded in package.json. A Git commit hash could not be verified because the project was downloaded as a ZIP archive rather than cloned with Git.

4. Main Features and User

Primary user: a general user who manages personal tasks.

- Add a task
- Edit an existing task
- Delete an existing task
- Mark a task as completed
- Mark a completed task as active again
- View all tasks
- View active tasks only
- View completed tasks only
- View the remaining-task counter

5. Application Rules / Acceptance Criteria

Rule ID	Rule / Acceptance Criterion
R-01	The user should be able to add a new task when valid, non-empty text is entered.
R-02	The application should not add a task when the task input is empty.
R-03	The user should be able to edit the text of an existing task.
R-04	The user should be able to delete an existing task.
R-05	The user should be able to mark an active task as completed.
R-06	The user should be able to mark a completed task as active again.
R-07	The All filter should display both active and completed tasks.
R-08	The Active filter should display only tasks that are not completed.
R-09	The Completed filter should display only completed tasks.
R-10	The remaining-task counter should show the number of active (incomplete) tasks.

6. Assumptions and Test Conditions

- The application is installed and running in a supported web browser.
- The user can access the task input, task list, filters, edit control, delete control, and completion checkbox.
- Each test starts with the exact task data stated in its preconditions; test data is reset between cases when necessary.
- A task is considered active when it has not been marked completed.
- The remaining-task counter represents the number of active (incomplete) tasks.
- An empty task is expected to be rejected; this is the expected rule to be checked by TC-02.

7. Test Cases

Instruction: The Actual Result field is intentionally left blank and every test case is marked Not run, as required by the lab task.

TC-01 — Add a valid task

Requirement	R-01
Preconditions	Application is running; All filter is selected; no tasks exist.
Test Data	Complete Lab Assignment
Steps	1. Enter “Complete Lab Assignment” in the new-task input field. 2. Click the Add button. 3. Inspect the task list and input field.

Expected Result	The task “Complete Lab Assignment” appears exactly once as an active task, and the input field is cleared.
Actual Result	
Status	Not run

TC-02 — Attempt to add an empty task

Requirement	R-02
Preconditions	Application is running; All filter is selected; no tasks exist.
Test Data	Empty input (no characters or spaces).
Steps	1. Leave the new-task input field empty. 2. Click the Add button. 3. Inspect the task list and remaining-task counter.
Expected Result	No task is added. The task list remains empty and the remaining-task counter remains 0.
Actual Result	
Status	Not run

TC-03 — Edit an existing task

Requirement	R-03
Preconditions	Application is running; one active task named “Old Assignment” exists.
Test Data	Existing task: Old Assignment; replacement text: Updated Assignment.
Steps	1. Locate “Old Assignment” in the task list. 2. Click its Edit control. 3. Replace the task text with “Updated Assignment”. 4. Save or confirm the edit using the control shown by the application. 5. Inspect the task list.
Expected Result	The task is displayed as “Updated Assignment”. The old text “Old Assignment” is no longer displayed as a separate task.
Actual Result	
Status	Not run

TC-04 — Delete an existing task

Requirement	R-04
Preconditions	Application is running; one active task named “Delete Me” exists.
Test Data	Delete Me

Steps	1. Locate the task “Delete Me”. 2. Click the Delete control for that task. 3. Inspect the task list.
Expected Result	The task “Delete Me” is removed from the task list.
Actual Result	
Status	Not run

TC-05 — Mark an active task as completed

Requirement	R-05
Preconditions	Application is running; one active task named “Testing Task” exists.
Test Data	Testing Task
Steps	1. Locate “Testing Task”. 2. Click its completion checkbox. 3. Inspect the task's visual status and the remaining-task counter.
Expected Result	The task is marked completed and is no longer counted as an active/remaining task.
Actual Result	
Status	Not run

TC-06 — Reactivate a completed task

Requirement	R-06
Preconditions	Application is running; one completed task named “Completed Task” exists.
Test Data	Completed Task
Steps	1. Locate the completed task “Completed Task”. 2. Click its completion checkbox again. 3. Inspect the task's visual status and the remaining-task counter.
Expected Result	The task changes back to active/incomplete status and is counted as a remaining task.
Actual Result	
Status	Not run

TC-07 — Display all tasks

Requirement	R-07
Preconditions	Application is running; two active tasks and one completed task exist.
Test Data	Active A; Active B; Completed C (marked completed).
Steps	1. Select the All filter. 2. Inspect the displayed task list.

	3. Count the visible tasks.
Expected Result	All three tasks (Active A, Active B, and Completed C) are displayed.
Actual Result	
Status	Not run

TC-08 — Display active tasks only

Requirement	R-08
Preconditions	Application is running; two active tasks and one completed task exist.
Test Data	Active A; Active B; Completed C (marked completed).
Steps	1. Select the Active filter. 2. Inspect the displayed task list.
Expected Result	Only Active A and Active B are displayed. Completed C is not displayed.
Actual Result	
Status	Not run

TC-09 — Display completed tasks only

Requirement	R-09
Preconditions	Application is running; two active tasks and one completed task exist.
Test Data	Active A; Active B; Completed C (marked completed).
Steps	1. Select the Completed filter. 2. Inspect the displayed task list.
Expected Result	Only Completed C is displayed. Active A and Active B are not displayed.
Actual Result	
Status	Not run

TC-10 — Verify the remaining-task counter

Requirement	R-10
Preconditions	Application is running; the task list contains two active tasks and one completed task.
Test Data	Active A; Active B; Completed C (marked completed).
Steps	1. Ensure Active A and Active B are incomplete. 2. Ensure Completed C is marked completed. 3. Inspect the remaining-task counter.
Expected Result	The remaining-task counter displays 2.
Actual Result	
Status	Not run

8. Conclusion

This report defines ten application rules and ten corresponding test cases for TodoMatic. The test cases cover task creation, invalid empty input, editing, deletion, completion, reactivation, filtering, and the remaining-task counter. Each test case is linked to a rule ID and includes preconditions, test data, ordered steps, and an observable expected result. Actual Result fields remain blank and all statuses remain Not run, as instructed.

9. Note on Initial Application Exploration

The application was downloaded from the specified GitHub repository and launched locally. During initial familiarization, the task-management controls were explored. Empty input was observed to create an empty task in that exploratory attempt; TC-02 therefore checks the expected rejection of empty input. This exploratory observation is not entered as a formal Actual Result because the instructor requires all formal test cases to remain Not run.